Simulating Random Graph Models in Python

Dissertation Presentation

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Why Study Random Graph Models?

Networks are everywhere:

- Social networks, the internet, neural networks, citation graphs
- All share structural properties that cannot be captured by deterministic models

Key structural questions:

- How do large networks form and evolve?
- Why do real-world networks have short average path lengths (*small-world property*)?
- Why do a small number of nodes attract the majority of connections (*hubs*)?

Goal of this Project

Understand the **mathematical foundations** of three principal random graph models and explore their theoretical structural properties through rigorous analysis and simulation.

The Erdős–Rényi Model $G(n, p)$

Definition

Let $n \in \mathbb{N}$ be the number of vertices and $p \in [0, 1]$. The random graph $G(n, p)$ is formed on vertex set $V = \{1, \ldots, n\}$ by including each possible edge independently with probability p.

Fundamental properties:

- **Expected edges:** $\mathbb{E}[|E|] = p\binom{n}{2}$
- **Degree distribution:** $d(v) \sim \mathrm{Binomial}(n-1,\ p)$, so $\mathbb{E}[d(v)] = p(n-1)$
- **Connectivity threshold:** $p_c = \dfrac{\ln n}{n}$

  For $p > p_c$, the graph is *almost surely* connected as $n \to \infty$; for $p < p_c$, isolated vertices persist almost surely.

Erdős–Rényi: Structural Interpretation

Phase transitions:

- Below p_c: the graph consists of many small, disconnected components.
- At $p = \frac{1}{n}$: a *giant component* of size $\Theta(n)$ emerges.
- Above p_c: a single connected component dominates with high probability.

Limitations as a real-world model:

- Degree distribution is **Binomial** (tightly concentrated around mean) — no heavy tails.
- **Low clustering:** the probability of two neighbours of v being connected is simply p — clustering coefficient $C \approx p \ll 1$ for sparse graphs.
- Does not replicate the high clustering observed in social or biological networks.

Key Takeaway

$G(n, p)$ gives sharp threshold theory but misses key real-network features: strong clustering and heavy-tailed degree distributions.

The Watts–Strogatz Small-World Model

Construction

1. Begin with a **regular ring lattice**: n vertices arranged in a circle, each connected to its k nearest neighbours (k even, $k \ll n$).
2. For each edge, **rewire** it with probability $\beta \in [0, 1]$ to a uniformly chosen random vertex (avoiding self-loops and duplicates).

Effect of the rewiring parameter β:

- $\beta = 0$: pure ring lattice — high clustering, large average path length.
- $\beta = 1$: effectively a random graph — low clustering, short path length.
- $0 < \beta \ll 1$: **small-world regime** — a few long-range shortcuts dramatically reduce average path length while clustering remains high.

Watts–Strogatz: Clustering and Path Length

Clustering coefficient (informally): the probability that two neighbours of a vertex are themselves connected — a measure of local cohesion.

Why clustering stays high for small β:

- Most edges remain in place; the local lattice structure is preserved.
- The fraction of rewired edges is small, so triangles are not destroyed.

Why average path length drops sharply:

- Even a small number of random shortcuts act as *bridges* between distant parts of the network.
- Reduces diameter from $O(n/k)$ on the lattice to $O(\log n)$.

The Small-World Phenomenon

High clustering (like a lattice) *and* short path lengths (like a random graph) can coexist — precisely the regime observed in real-world networks.

The Barabási–Albert Preferential Attachment Model

Construction

1. Begin with a small initial connected graph on m_0 vertices.
2. At each time step, add one new vertex with $m \leq m_0$ edges.
3. Each new edge connects to an existing vertex i with probability proportional to i's current degree:

$$P(i) = \frac{d_i}{\sum_j d_j}$$

Intuition: *"The rich get richer."*

High-degree vertices attract more connections — a positive feedback mechanism.

Barabási–Albert: Scale-Free Networks

Power-law degree distribution:

$$P(k) \sim k^{-3}, \qquad k \to \infty$$

This is derived rigorously via a mean-field or master-equation approach and holds asymptotically as the network grows.

What power-law implies structurally:

- **Heavy tail:** A small number of vertices (*hubs*) accumulate a disproportionately large share of all edges.
- **Scale-free:** No characteristic degree scale exists — the distribution has no finite variance in the limit.
- **Robustness and vulnerability:** The network is highly robust to random failures (most nodes have low degree) but fragile under targeted attacks on hubs.

Key Takeaway

Preferential attachment explains heavy-tailed degrees and hubs seen in web and citation networks.

Comparative Structural Properties

	Erdős–Rényi	**Watts–Strogatz**	**Barabási–Albert**
Mechanism	Independent edge probability p	Rewiring a ring lattice	Growth + preferential attachment
Degree dist.	Binomial$(n-1, p)$	Approx. Poisson for small β	Power law $P(k) \sim k^{-3}$
Clustering	Low ($\approx p$)	High for small β	Low–moderate
Hubs	None	None	Present (heavy tail)
Typical uses	Connectivity thresholds, percolation	Social, neural networks	Web, citation, scale-free nets

Theoretical Implications and Conclusions

Theoretical implications:

- **ER:** Provides exact threshold phenomena via probabilistic methods, but is too homogeneous for most real networks.
- **WS:** Demonstrates that *local structure* and *global efficiency* are not mutually exclusive — a key insight for network science.
- **BA:** Shows that *growth* and *feedback* alone are sufficient to generate empirically observed heavy-tailed distributions.

Limitations:

- WS does not produce power-law degree distributions.
- BA has low clustering — hybrid models (e.g., Holme–Kim) address this.
- All three assume *undirected, unweighted* graphs.

Summary

Each model captures a different structural mechanism. Together they motivate richer models for real-world complex networks.